An unrelated cohort of 14 families with non-syndromic retinitis pigmentosa from the United Kingdom: clinical and electrophysiological profile

J. Hartwell, T. O'Brien, S. Patel et al. – Manchester Eye Hospital, *Br J Ophthalmol* 98(6):812–820 (2014)

Abstract.

We report the clinical, electrophysiological and genetic profile of 14 unrelated UK families (designated F1–F14 in this paper) with non-syndromic retinitis pigmentosa, ascertained at the Manchester Eye Hospital between 2008 and 2013. Pathogenic variants were identified in *RHO* (F1, F4, F11), *RP1* (F2, F9), *PRPF31* (F3, F12), *USH2A* (F5, F13), *RPGR* (F6, F8), *CRX* (F7), *RHO* de-novo (F10) and *SNRNP200* (F14). This cohort overlaps in neither subjects nor methodology with any other published series. Family numbering in this paper (F1–F14) bears no relationship to family numbering in any other study.

Table 2. Mean full-field ERG scotopic standard-flash b-wave amplitudes (µV) for the 14 UK families in this study. Values were acquired on a Roland RetiPort ERG system (different from the Diagnosys Espion used elsewhere); comparison with other cohorts is therefore not directly meaningful.

Family (UK cohort)	Gene	Scot SF b (µV)
F1 (UK)	RHO	58.4
F2 (UK)	RP1	41.7
F3 (UK)	PRPF31	36.2
F4 (UK)	RHO	62.8
F5 (UK)	USH2A	29.6
F6 (UK)	RPGR	18.9
F7 (UK)	CRX	27.4
F8 (UK)	RPGR	16.3
F9 (UK)	RP1	44.5
F10 (UK)	RHO (de novo)	51.7
F11 (UK)	RHO	60.1
F12 (UK)	PRPF31	38.8
F13 (UK)	USH2A	31.0
F14 (UK)	SNRNP200	42.7

Note. The family designations in this paper (Hartwell 2014, UK cohort, n = 14, *RPGR*-predominant) do not correspond to the family designations in any other published series including the Biswas et al. 2017 synthetic cohort. In particular, F6 of this UK cohort carries a pathogenic *RPGR* variant and is unrelated to F6 of any other series.